PCA of ToF-SIMS spectra

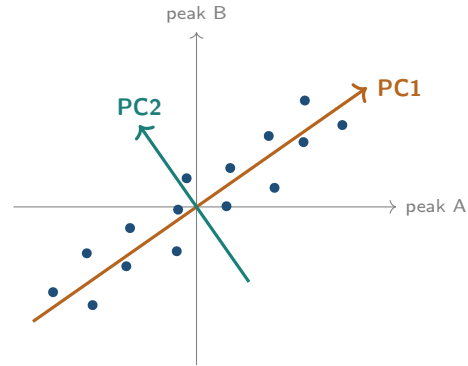
All-peak PCA and Selected-peak PCA

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What PCA does

- A sample is a spectrum: hundreds of numbers (here **901 bins**).
- PCA finds a few new axes, **PC1**, **PC2**, . . . , that keep as much of the difference between samples as possible.
- Samples with similar spectra end up **close together** on the map.

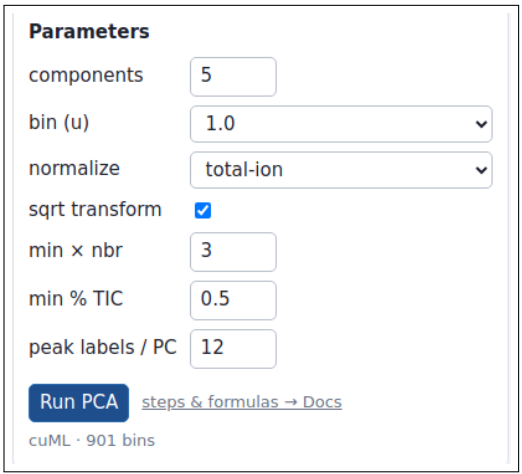


PC1 follows the longest direction of the cloud

Two things to read

Scores: one dot per sample.
Loadings: which peaks pull each axis.

Transformations: the settings in the interface



Statistical methods tab, parameters panel

- `bin (u)`: width of each m/z bin. One bin = one variable.
- `normalize = total-ion`: divide each spectrum by its own total, $x / \sum x$. Removes differences in total signal.
- `sqrt transform`: take $\sqrt{x}$. Shrinks the big peaks and evens out counting noise.
- **Mean centring** is always applied: subtract the average spectrum.
- `components`: how many PCs to compute.

normalise $\rightarrow \sqrt{\ } \rightarrow$ **centre** $\rightarrow$ **PCA**

Two rounds of PCA (Yang et al. 2019)

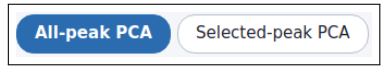


- All-peak PCA: strong matrix peaks (in Yang's study, culture-medium salts) dominate the result.
- Selected-peak PCA: peaks dominated by the medium are dropped, characteristic ones are kept. The same samples separate, and the **loadings are easier to interpret**.
- Rule of thumb for round 2 (Fu et al. 2018): a peak is at least **3×** its neighbours and holds more than **0.5%** of the ions. These are `min × nbr` and `min % TIC` in the interface.

Yang et al., *Biointerphases* 14, 051004 (2019). Fu et al., *Atmos. Environ.* 191, 36 (2018).

In the interface: from All-peak to Selected-peak

1. Pick subsamples and labels, click `Run PCA` on the **All-peak PCA** tab.
2. The `Peak list` ranks every bin by its influence on the PCs. The 40 strongest are pre-ticked. `pass` marks bins that meet the 3× and 0.5% rule.
3. Tick or untick peaks.
4. Open the **Selected-peak PCA** tab: the same transformations, only on the ticked peaks.



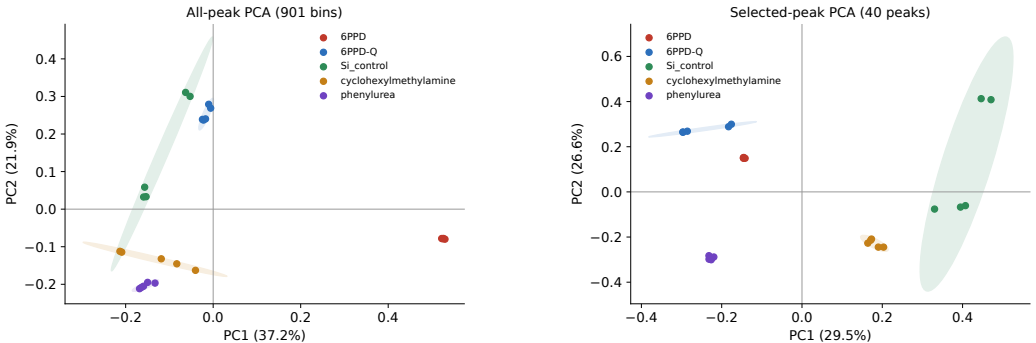
search name or m/z | passing | ticked | tick shown | untick shown | ⬇ CSV | 901 of 901 features (fir

✓	m/z	window	counts	% ions	× nbr	max |p| ▼
☑	22.50	22.00–23.00	6,094,091	0.51	5.1 pass	0.375
☑	62.50	62.00–63.00	3,681,684	0.31	1.3	0.292
☑	20.50	20.00–21.00	1,995,729	0.17	2.5	0.243
☑	48.50	48.00–49.00	3,385,589	0.28	3.4	0.227
☑	38.50	38.00–39.00	6,654,254	0.56	0.8	0.217
☑	36.50	36.00–37.00	1,283,792	0.11	2.6	0.194
☑	219.50	219.00–220.00	6,122,300	0.52	1.7	0.184
☑	18.50	18.00–19.00	4,034,931	0.34	5.1	0.184
☑	73.50	73.00–74.00	4,201,031	0.35	2.2	0.178

Peak list: ticked peaks, m/z, share of ions, × neighbour, pass, loadings

Not in the tool: removal of PDMS and Bi primary ions, and Yang's medium-overlay filter.

Real data: the two PCAs side by side



- 24 positive-polarity subsamples, five groups. Same transformations in both.
- PC1 + PC2 explain **59.1%** (901 bins) and **56.1%** (40 peaks).
- The 40 peaks are picked from the same data, so the separation is **descriptive**, not a test.